

The First Law of Thermodynamics The Microscopic Model As you can see here, we are moving

[H]

[width=0.6] /PhyAI/chapter1/1_0.png Our hypothesis model of gas and crystal But why do I call these particles "imaginary

[width=0.7] /PhyAI/chapter1/1_1.png Definition of Volume Now you might wonder if you mix 2 types of completely different

[width=0.7] /PhyAI/chapter1/1_2.png Mixing H_2 and He together Reaching the equilibrium state means the density of gas

Unlike volume, **pressure** is very easy concept to define: [The First Definition of Pressure] Pressure is the force applied

²), or (Pa)

One classic example of pressure is pressing your hand against a small piston filled with gas. If you press hard and fast enough, you can break the piston. [width=0.6] /PhyAI/chapter1/13.png Pressing a small piston After understanding what volume and pressure are, now it is

This section does not have much to discuss beyond introducing an important law of gases and some new constants. As you can see, the model is oversimplified. [width=0.5] /PhyAI/chapter1/15.png An oversimplified model of an ideal gas Since the system of 10^{23} particles is impossible to simulate, I oversimplified the model. My oversimplified model of an ideal gas contains only 1 particle. Its speed never changes.

Piston is perfectly smooth, so collisions are always **elastic**.

Directly applying Definition yields: $\overline{P}_{x, \text{on piston}} = \frac{\overline{F}_{x, \text{on piston}}}{A} = -\frac{\overline{F}_{x, \text{on particle}}}{A} (\text{Newton's Third Law})$

$$= -\frac{m \frac{\Delta v}{\Delta t}}{A} = -\frac{m \frac{(-v_x) - v_x}{2L/v_x}}{A} = \frac{mv_x^2}{V}$$

If our piston contains N particles, then the total pressure on piston by x -axis is:

Because N is extremely large, and the gas is homogeneous:

The average velocity of the particles in the x, y and z directions is the same, we obtain:

The average velocity of N particles in three-dimensional space is defined as:

The average translational kinetic energy is then:

We can simply explain the phenomenon of thermal expansion using the previous result. If the temperature increases, the average velocity of the particles increases, and the pressure increases.

$$\text{Moving along } x\text{-axis: } \overline{K}_x = \frac{1}{2} m \overline{v_x^2} = \frac{k_B T}{2}$$

$$\text{Moving along } y\text{-axis: } \overline{K}_y = \frac{1}{2} m \overline{v_y^2} = \frac{k_B T}{2}$$

$$\text{Moving along } z\text{-axis: } \overline{K}_z = \frac{1}{2} m \overline{v_z^2} = \frac{k_B T}{2}$$

Other degrees of freedom include rotational motion and elastic potential energy (as stored in a structure like a spring), and vibrational energy.

The final problem is determining the value of f . For example, if our gas is monatomic, then we can easily conclude that $f = 3$.

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$$\text{Moving along } z\text{-axis: } \overline{K}_z = \frac{1}{2} m \overline{v_z^2} = \frac{k_B T}{2}$$

$$\text{Rotating around } x\text{-axis: } \overline{K}_{x, \text{rotate}} = \frac{1}{2} I \omega_x^2 = \frac{k_B T}{2}$$

$$\text{Rotating around } y\text{-axis: } \overline{K}_{y, \text{rotate}} = \frac{1}{2} I \omega_y^2 = \frac{k_B T}{2}$$

Most polyatomic molecules can rotate about all three axes, so if our gas is polyatomic, then $f = 6$. In the case of solid, $f = 6$. [width=0.8] /PhyAI/chapter1/16.png Degrees of freedom illustration Counting degrees of freedom of liquid is more complicated. [width=0.7] /PhyAI/chapter1/17.png Heat Q is converting to work W to push the piston away I think we are all familiar with

Both work W and heat Q are 2 types **energy**; but they have a slight but subtle difference:

Heat is defined as any **spontaneously flow of energy** from one object to another. As we discussed in the previous chapter, work is defined as energy transferred by a force acting through a distance.